

GrounUp Enterprise Specification

Phase 22 - Final Reconciliation, Enterprise Lock & Developer/AI Execution Package

Version 1.0.0 | Review-ready | Enterprise baseline NOT LOCKED

1. Mission

Turn the consolidated Phase 00-21 specification into the authoritative execution, acceptance and lock package for building GrounUp with human developers and AI-assisted coding systems.

2. Source Preservation

Phase 22 preserves 4,785 consolidated source requirements from Phase 21. Source wording is not silently rewritten. Findings remain governed review items until an authorized owner records a disposition.

3. Final Reconciliation

Phase 21 identified 278 structural findings. Phase 22 carries them into a final disposition register and requires semantic cross-phase review for architecture, business rules, data contracts, APIs/events, security, privacy, AI governance and testing.

4. Implementation Baseline

The approved dependency waves become executable program epics. Every development work item references requirement IDs, mapped tests and relevant phase artifacts.

5. Developer & AI Execution Contract

AI and human developers operate under the same provenance, testing, tenant-isolation, security, versioning, audit, migration, observability and change-control rules. AI cannot silently invent missing governed decisions or approve protected gates.

6. Acceptance

Definition of done includes implementation, tests, documentation, observability, security, traceability, acceptance and release readiness. Production additionally requires Phase 20 readiness controls.

7. Enterprise Lock

GES v1.0 remains NOT LOCKED until all mandatory Phase 22 gates are approved and blocking findings are closed or formally excepted. Lock is an explicit configuration-management action, not an automatic result of documentation generation.

8. Post-Lock Governance

After lock, specification changes require a governed change request, impact analysis, approvals, version update, regression scope and updated manifest/traceability.

Final Approval Gates

Gate	Owner	Status	Evidence
Scope & Phase Catalog	Product Owner	Pending	Phase 00-22 scope/version catalog
Requirements Reconciliation	Product + Architecture	Pending	Final Requirements Register + dispositions
Architecture/Data/API	Architecture + Data	Pending	Canonical contracts approved
Security & Privacy	Security	Pending	Security/privacy acceptance evidence
AI Governance	AI Governance	Pending	AI evaluations/guardrails/ownership
Testing & Traceability	QA	Pending	Coverage + exception evidence
Business/UAT	Business Owners	Pending	UAT acceptance
Production Readiness	DevOps/SRE	Pending	Phase 20 readiness evidence
Operations/Support	Service Management	Pending	Runbooks, SLAs, support ownership
Final Manifest	Configuration Manager	Pending	Hash manifest + approved baseline
GES v1.0 Enterprise Lock	Configuration Manager	Proposed	All prior gates approved

Developer / AI Execution Controls

P22-EXEC-001 - Requirement Provenance: Every code/change task cites authoritative requirement IDs and source phase. Enforcement: Block merge if absent.

P22-EXEC-002 - No Silent Assumptions: Missing business rules/formulas/permissions become governed gaps. Enforcement: Do not invent.

P22-EXEC-003 - Test With Implementation: Mapped automated/manual acceptance evidence accompanies implementation. Enforcement: Block completion without test evidence.

P22-EXEC-004 - Tenant Isolation: All SaaS data, config, entitlements and branding remain tenant scoped. Enforcement: Security gate.

P22-EXEC-005 - API Contract Discipline: Approved API/event schemas are versioned compatibility boundaries. Enforcement: Breaking changes require new version/change approval.

P22-EXEC-006 - Data Migration Discipline: Schema/data changes include migration, reconciliation and rollback. Enforcement: Deployment gate.

P22-EXEC-007 - AI Governance: AI models/prompts/tools require versioning, evaluation, guardrails, monitoring and human policy. Enforcement: AI gate.

P22-EXEC-008 - Observability: Production features emit governed logs/metrics/traces and audit events. Enforcement: SRE gate.

P22-EXEC-009 - Security by Default: Least privilege, secrets management, encryption and vulnerability gates apply. Enforcement: Security gate.

P22-EXEC-010 - Feature Flags: Incomplete/high-risk releases use governed feature flags when appropriate. Enforcement: Release control.

P22-EXEC-011 - Rollback Ready: Every production-impacting release has verified rollback/recovery behavior. Enforcement: Production gate.

P22-EXEC-012 - Traceability Update: Accepted changes update requirement-test-code-release traceability. Enforcement: Definition of done.

P22-EXEC-013 - Human Approval Boundaries: AI cannot approve legal, financial, security, safety, contractual or enterprise-lock decisions. Enforcement: Governance gate.

P22-EXEC-014 - Canonical Libraries: Shared services/materials/resources/rates/assemblies use governed master-library identifiers. Enforcement: Data governance gate.

P22-EXEC-015 - Audit Immutability: Approval, release, pricing, commercial and security audit records preserve history. Enforcement: Audit gate.

P22-EXEC-016 - Environment Promotion: Artifacts/configuration promote through approved environments; no ad hoc production mutation. Enforcement: DevOps gate.

P22-EXEC-017 - White Label Governance: Branding/customization is configuration-driven and cannot fork core business logic silently. Enforcement: SaaS architecture gate.

P22-EXEC-018 - Performance Acceptance: Critical workflows meet approved latency/throughput/capacity targets.
Enforcement: Performance gate.

P22-EXEC-019 - Accessibility Acceptance: User-facing surfaces meet approved accessibility criteria.
Enforcement: UX/QA gate.

P22-EXEC-020 - DR Acceptance: Backup/restore and RTO/RPO recovery evidence exists for critical services.
Enforcement: DR gate.

P22-EXEC-021 - Commercial Calculation Controls: Estimate/job cost/change/claim/payment calculations retain formula/version/source provenance. Enforcement: Finance/commercial gate.

P22-EXEC-022 - Construction AI Evidence: Plan/spec/takeoff AI outputs retain source-page/location/confidence/review evidence. Enforcement: AI/construction QA gate.

P22-EXEC-023 - Change Control: Post-lock changes require change request, impact analysis, approvals, version update and regression scope. Enforcement: Baseline control.

P22-EXEC-024 - Definition of Done: Requirement implemented, tested, documented, observable, secure, traceable, accepted and releasable. Enforcement: Completion gate.